

Geophysical Research Letters®

RESEARCH LETTER

10.1029/2025GL117138

Key Points:

- Slip parallel to the b direction in olivine is said to be rare but here we describe a sample with 17% of such dislocations
- Results from Weighted Burgers Vector calculations on Electron Backscatter Diffraction maps, confirmed by Transmission Electron Microscopy
- Our approach could be applied to other rocks and minerals using just Electron Backscatter Diffraction, with Transmission Electron Microscopy an optional follow-up

Supporting Information:

Supporting Information may be found in the online version of this article.

Correspondence to:

J. Wheeler,
johnwh@liverpool.ac.uk

Citation:

Wheeler, J., Hunt, S., Eggeman, A., Donoghue, J., Gholinia, A., Li, Y., et al. (2025). Olivine deformation: To b slip or not to b slip, that is the question. *Geophysical Research Letters*, 52, e2025GL117138. <https://doi.org/10.1029/2025GL117138>

Received 30 MAY 2025

Accepted 24 NOV 2025

Author Contributions:

Conceptualization: J. Wheeler, S. Hunt, A. Eggeman, A. Gholinia, Y. Li, E. Tillotson

Funding acquisition: J. Wheeler, S. Hunt, S. J. Haigh

Investigation: S. Hunt, A. Eggeman, A. Gholinia, Y. Li, E. Tillotson

Methodology: J. Wheeler, S. Hunt, A. Eggeman, A. Gholinia, Y. Li, E. Tillotson

Project administration: J. Wheeler

Software: J. Wheeler

Visualization: E. Tillotson

Writing – original draft: J. Wheeler

Writing – review & editing: J. Wheeler, S. Hunt, A. Eggeman, A. Gholinia, Y. Li, E. Tillotson, S. J. Haigh

© 2025. The Author(s).

This is an open access article under the terms of the [Creative Commons Attribution License](#), which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

Olivine Deformation: To B Slip or Not to B Slip, That Is the Question

J. Wheeler¹ , S. Hunt², A. Eggeman² , J. Donoghue², A. Gholinia² , Y. Li², E. Tillotson² , and S. J. Haigh²

¹Department Earth, Ocean and Ecological Sciences, University of Liverpool, Liverpool, UK, ²Department of Materials, University of Manchester, Manchester, UK

Abstract When deformed by dislocation creep the dominant slip (Burgers) vectors of olivine dislocations are parallel to [100] or [001]. Dislocations with an [010] Burgers vector component (b dislocations) have been recorded rarely. Here we show an experimentally deformed olivine sample has a substantial population (17%) of b dislocations. Electron Backscatter Diffraction maps of crystal orientations provided information on dislocations from the orientation gradients. Maps show the b dislocations form subgrain walls like those formed by other dislocation types and are interpreted to form similarly by glide and climb, so b dislocations are mobile. To confirm our approach, we used Electron Backscatter Diffraction maps to select an area for Transmission Electron Microscopy imaging, down to an atomic scale image of a b dislocation. Our sample was deformed within range of subduction zone conditions; our approach can be used to investigate the scale and conditions of b slip in the mantle more widely.

Plain Language Summary Below the relatively rigid tectonic plates, the top 400 km of the Earth (the upper mantle) is solid but in constant slow movement linked to the movements of plates. To understand such movements, we need to understand how the main mineral in the upper mantle, olivine, deforms. Like all minerals, olivine has a regular repeating atomic structure. Deformation is enabled as linear imperfections (dislocations) move through each crystal and enable slip parallel to various specific directions in specific atomic planes. In one direction (called b) the atomic repeat distance is large, and it is thought that b dislocations are rare and/or cannot move. Here we use a Scanning Electron Microscopy method to make maps of crystal orientations on the scale of microns. We analyzed variations in crystal orientation and found b dislocations. To confirm their presence, we used Transmission Electron Microscopy giving images down to the scale of atoms. We suggest b dislocations may be more widespread than thought, thus improving our understanding of mantle deformation. Their presence may be influenced by pressure, temperature and/or stress levels—so measuring b dislocations in natural samples may help deduce what depth they came from, and the conditions of deformation.

1. Introduction

Olivine is the dominant mineral in the convecting upper mantle, so it is essential to understand its deformation to understand how the mantle deforms. Experiments have measured the stress-strain relationships of olivine as a function of grain-size, pressure, temperature and strain-rate. The resulting flow-laws have been used to model (e.g.,) mantle deformation (Blackman & Kendall, 2002), convection (Becker, 2006) and glacial isostatic rebound (Whitehouse, 2018). The link between experiments and nature requires extrapolation of flow laws determined on laboratory time scales and is strengthened by understanding how deformation links to observable microstructures. Dislocation creep plays a key role in upper mantle deformation (Ohuchi et al., 2015) so we need to understand the fingerprints it leaves in microstructures, and the influences of ambient conditions on dislocation activity. Crystallographic preferred orientations (CPOs) that developed during experiments are used to interpret the dislocation slip system activities and seismic anisotropies observed in the Earth's upper mantle and to relate experiments to nature. Understanding the slip systems underpins models for rheology and CPO development; they can also give information on stress and water activity (H. Jung et al., 2006). However the link between CPO and slip systems is not direct—"easy slip systems ... control the LPO development" (H. Jung et al., 2006) but Transmission Electron Microscopy (TEM) is needed to confirm dislocation types which means very restricted areas are investigated.

Olivine unit cell axes have lengths of $a = 4.76$, $b = 10.22$ and $c = 5.99$ Å, so [010] dislocations are expected to have long Burgers vectors and high energies. They may therefore dissociate and are argued to be difficult to form and move. Experiments on single crystals (Table 1) at low pressure support the dominance of slip parallel to [100]

Table 1
Reports of b Dislocations, or Their Absence, in Order of Increasing Confining Pressure

Pressure (GPa)	Stress (GPa)	T (°C)	Study indicating presence or absence of b
<0.18	<0.1	1,150–1,600	No b: single crystals (Durham & Goetze, 1977)
0.3	0.3	1,050–1,250	No b: single crystals (Tielke et al., 2017)
1	1.28	800	(100)[010] slip: single crystals (Phakey et al., 1972), described as “suspect” (Boland & Buiskool Toxopeus, 1977)
1?	?	1,000	Found “fortuitously” in shallow oceanic mantle mylonitized peridotite (Demouchy et al., 2024)
1.8	?	1,100	Dissociated [010] dislocations from a peridotite originating in the upper mantle (Fujino et al., 1993)
<1 then 1.5–2.1	?	800–900 then 650–700	Natural CPO from granulite facies lherzolite with later eclogite facies event (S. Jung et al., 2014)
2	0.5	850 ± 50	<i>This study</i> : 17% of dislocations are b
3	?	800 ± 175	Natural CPO that is best explained by [010] slip (Skemer et al., 2006)
5	0.1	1,400	[010] deduced from subgrain wall geometry in single crystal (Ando et al., 1997)
5.2	internal	1,100	Ni ₂ SiO ₄ olivine partly transformed to spinel (Boland & Liebermann, 1983)

Note. Stress in natural samples is not constrained; stress in the Ni₂SiO₄ experiment was generated internally by phase transformation and not constrained. Pressure is not stated in Demouchy et al. (2024) so we give a rough figure based on the geological setting.

or [001]. This means that the well-known olivine slip systems together do not satisfy the von Mises criterion (i.e., they cannot accommodate an arbitrary imposed strain) and other deformation processes must be sought (Thieme et al., 2018). Table 1 summarizes observations or indirect deductions of b dislocations, ranked by pressure since Ando et al. (1997) suggests this has an effect. Their apparent rarity leads to a view that b dislocations are not a significant factor in olivine deformation but here we show that view needs revision.

Direct observation of b dislocations has to date been via TEM, which means that only small areas can be explored, and relative abundances are hard to quantify—b dislocations may be discovered “fortuitously” (Table 1). Here we precede such investigations with an analysis of Electron Backscatter Diffraction (EBSD) maps, extracting information on dislocations from intracrystalline orientation gradients (Wheeler et al., 2024). That method, on its own, provides key information but we used our results as a guide for confirmatory TEM work. We describe our methods first, then give our results.

2. Experimental and Analytic Methods

2.1. Experimental Details

Sample SiO₃₈ was a composite of San Carlos olivine powder with a 20- to 50-μm grain size and silica glass (separated by Pt foils), originally designed to study silica polymorph strengths, with olivine as a reference (Hunt et al., 2019). The sample was annealed in a DT-cup (Hunt et al., 2014) at 850 ± 50°C and 8.8 GPa and the glass transformed to coesite. The volume reduction caused a pressure drop to ~2 GPa prior to the samples being deformed by uniaxial compression. The deformation was applied in three stages with strain rates 2.1, 8.2 and 8.4 × 10^{−6}/s; the total strain at each step is 1.0%, 12.8% and 10.2% with corresponding differential stresses of 0.51, 0.93 and 0.81 GPa. Although outside its stability field the coesite persisted metastably, analogous to how stishovite can persist metastably at low P below at 800°C (Zhang et al., 1996).

We describe the analytic methods in order of increasing magnification and provide additional details in Supporting Information S1.

2.2. Scanning Electron Microscopy (SEM) Analysis

We imaged the sample using BSE so the image gray scales showing atomic number contrast and black cracks. EBSD then provided maps of crystal orientations (Prior et al., 2009). To visualize orientation, Inverse Pole Figure (IPF) maps were calculated from the Euler angles in which the X axis on the map is displayed as a color relative to the crystal axes (see key), so that variations in crystal orientation appear as varying colors.

The Weighted Burgers Vector (WBV) method was used to calculate information related to the Burgers vectors and abundances of different types of dislocation, using gradients in orientation (Wheeler et al., 2024). The method captures Geometrically Necessary Dislocations (GNDs) (Ashby, 1970), that is, those that give rise to lattice curvature (orientation variations) on the scale of the step size. The vector W is a sum of Burgers vectors of GNDs, weighted by the actual dislocation density of each type of GND and by the angle between the dislocation lines and the map. If we have N types of dislocation then W depends on density ρ , Burgers vector b and unit line vector l for each type (labeled N) of dislocation. Specifically, using subscript notation,

$$W_i = \sum_N \rho^{(N)} b_i^{(N)} l_3^{(N)} = \sum_N \left[\rho^{(N)} l_3^{(N)} \right] b_i^{(N)}$$

where W is the WBV, and the subscript “3” refers to the z direction, perpendicular to the map. Consequently, l_3 is the component of $[100]$ dislocation line vector perpendicular to the map: it varies from 0 (lines parallel to map) to 1 (lines perpendicular to map). For a given dislocation density, lines at a high angle to the map are favored (large l_3), explaining why the word “Weighted” is used in the method description. The terms in square brackets are scalars, so that WBV is a linear combination of Burgers vectors—this is one reason why it is a useful quantity. There may be more than one way to decompose W into specific Burgers vectors, but if W includes for example, $[010]$ there must be at least one type of GND with a Burgers vector with a $[010]$ component: the calculation cannot lead to “phantom” directions. The WBV direction (in crystal coordinates) is colored coded in maps using the IPF key.

A subarea was remapped by EBSD using a method which improves orientation accuracy to about 0.1° and consequently for the displayed smaller area map the WBV direction is accurate to $<10^\circ$.

2.3. Wafer for TEM

The EBSD data yielded key information on the presence of b dislocations, and to confirm this we undertook TEM. Guided by the WBV maps we identified an area with significant $[010]$ component to the GNDs. A linear structure was thought to be a subgrain wall, but we could not be sure in advance what type. We chose to extract a wafer (lamella) including the linear structure for TEM subparallel to the map surface so that we could correlate SEM and TEM images. FIB milling was used to excavate two sloping trenches which intersected below the surface, shown in the secondary electron (SE) image Figure 1e. After liftout the $15 \times 5 \mu\text{m}$ wafer was attached to a TEM grid and thinned to electron transparency.

2.4. TEM Images Above Atomic Scale

We give a simple outline of the meaning of the TEM images here, with more detail in Supporting Information S1.

We made a stitched Bright Field TEM image, which shows qualitative orientation contrasts, of the whole FIB liftout region. By comparing SE, BSE and EBSD images with this map, a linear feature running through the left side was identified as the dislocation wall which includes b dislocations from the WBV analysis. Further analysis using Scanning Precession Electron Diffraction (SPED) provided a set of diffraction patterns. From each pattern, the intensity of the unscattered beam was used to define an individual pixel gray scale on a synthetic map at the point where the pattern as collected, giving a Virtual Bright Field (VBF) image. Each diffraction pattern was also indexed to give crystal orientation, described by three Euler angles as it would be in EBSD. From these orientations we calculated misorientation axes relative to the starting orientation along a transect, marked as arrows, omitting those for which misorientation was $<1^\circ$ as angular errors would be significant.

Low-Angle Annular Dark field (LAADF)-STEM images were used to home in on further structures of interest. An annular detector is used, meaning that only scattered electrons are detected—the main beam carries on through the hole in the middle of the detector. The bright parts are where local distortions around dislocations favor scattering. Scattering angles are low so the contrast is dominated by diffraction effects.

2.5. TEM Images at Atomic Scale

We took ADF-STEM images in which the annular detector is set so that it collects electrons scattered through larger angles than for the ADF imaging discussed above. These electrons are mainly incoherently scattered, not

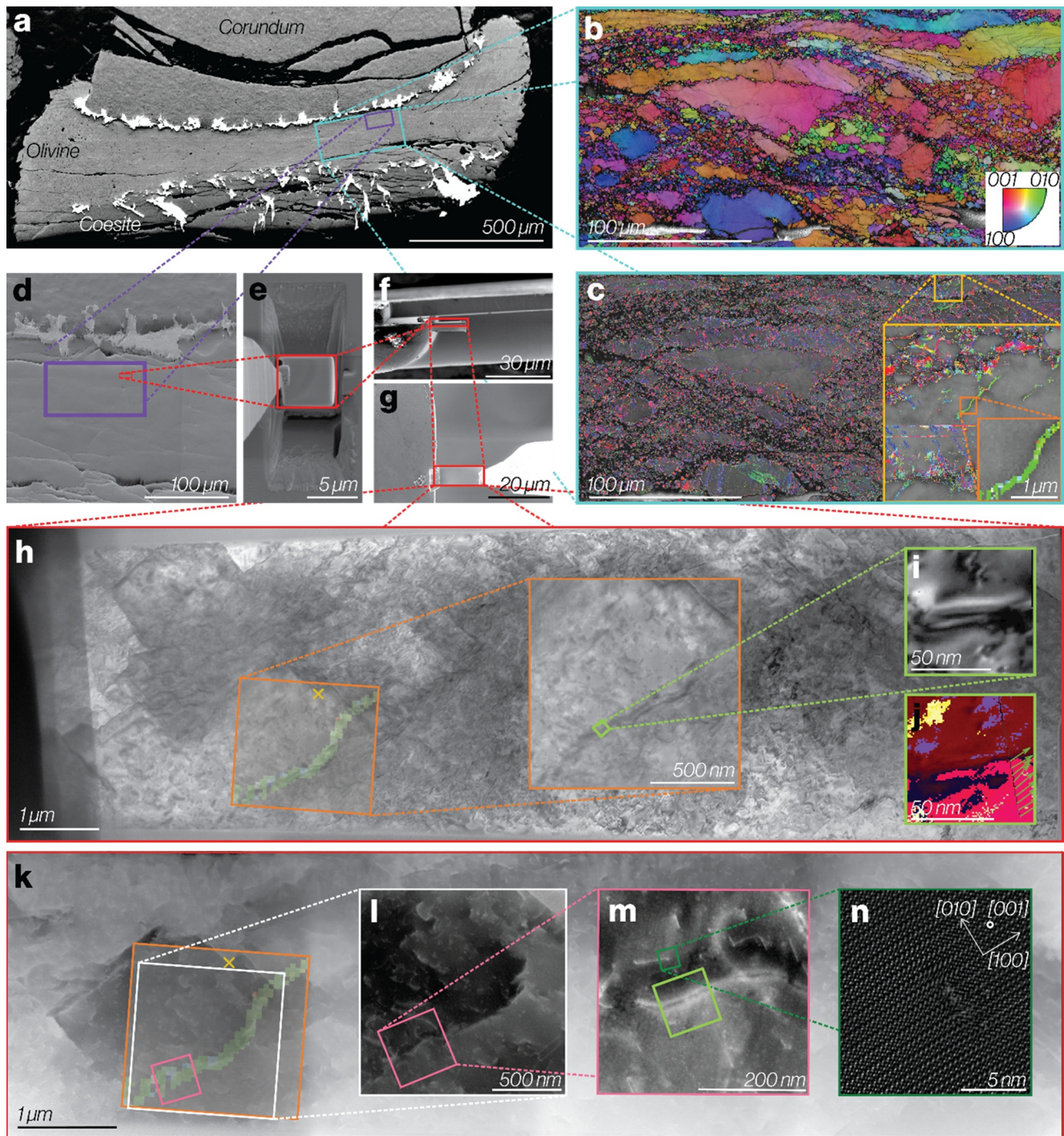


Figure 1. Electron microscopy characterization of b dislocations in olivine. (a) Low magnification backscatter electron image of highly deformed olivine with electron backscatter diffraction (EBSD) map, (b) from the blue rectangle shown in (a), color coded according to Z direction in crystal coordinates. (c) Accompanying weighted Burgers vector (WBV) map where green pixels (color coded by WBV direction using key shown in (b)) correspond with b dislocations. (d) Low magnification scanning electron microscope (SEM) image with the location of the EBSD map highlighted in purple, corresponding to that shown in (a). (e–g) Focused ion beam (FIB) sample preparation of plan-view lamella (red) showing liftout, attachment to transmission electron microscopy (TEM) grid and thinning to electron transparency, respectively. (h) Panoramic bright field image (BF) with magnified WBV map overlaid showing presence of b dislocation. (i) Virtual BF image acquired from the green square highlighted in (h) with (j) 4D-scanning transmission electron microscopy (STEM) orientation map color coded by Euler angles. The green arrows indicate the misorientation axes relative to the starting orientation along a transect (black line). (k) Panoramic LAADF image, corresponding with (h) with subsequent increases in magnification (l) and (m) related and color-coded through dashed lines. The gold marker shows the same location in both panoramic micrographs. (n) atomic-resolution ADF-STEM image.

diffracted, so images can—in a very simplified way—be regarded as direct images of atoms which lie within a few nm depth of field, an “optical section” (Yang et al., 2015). High Z atoms are brighter, and atoms directly behind each other add their brightnesses. Hence the brightest sharpest points are high Z atomic columns parallel to the beam; dimmer points may be lower Z columns parallel to the beam, and dimmer smeared-out shapes may be oblique columns. However, for these imaging conditions there are still diffraction effects in the ADF signal so the images were simulated by first creating atomic positions using CrystalMaker (Palmer, 2014) and then simulating the ADF image using a “multislice” method which takes full account of the wave optics (Kirkland, 2020).

We obtained an image with a defect. To interpret the simple periodic part of the image we overlaid the atomic positions calculated for 2×2 olivine unit cells viewed down [001], distinguishing M1 and M2 sites. To interpret the defect, we drew overlays of atomic plane traces by hand, to highlight offsets across that linear region. We also drew Burgers circuits to detect any extra half planes which would indicate edge dislocations.

For more detailed understanding we generated atomic models for displacements around dislocations using standard equations (Hirth & Lothe, 1982); these are not detailed models of core structure but assisted in understanding how and where columns of M1 atoms normally parallel to [001] would be deflected around various types of dislocation. We omitted M2 sites because these are almost invisible in the simulated ADF image due primarily to destructive interference (see Supporting Information S1 for details). We were mindful that, if the optical section was near the wafer surface, the free surface would modify stresses and displacements. Surface effects generally induce an elastic twist in screw dislocations (Eshelby, 1958; Eshelby & Stroh, 1951; Yoffe, 1961) and so we included a rotation in the vicinity of the screw dislocation in some models, to get an idea of how any elastic twist would modify the structure and the image.

The final stage of simulation was to take the atomic positions and mimic how they would look in an optical section. We assigned each atom in a narrow depth range (the optical section) a brightness in accord with its depth position, then generated a view down the electron beam direction, adding the brightnesses.

3. Results Concerning b Dislocations

3.1. Finding b Dislocations Using EBSD

The sample, in a backscatter electron image (Figure 1a), is separated from a corundum piston (above) and a coesite sample (below) by the remnants of a Pt foil (bright). EBSD data were collected from the thinnest, most highly deformed olivine region. Plotted as an Inverse Pole Figure (IPF) map (Figure 1b), the very distinct colors indicate originally separate powder grains with different orientations, now severely elongate and strained. Subtle color variations indicate intragrain distortion and hence the presence of lattice distortion and GNDs. These were quantified using the WBV method giving maps (Figure 1c) with the same IPF color scheme but now indicating the orientation of the WBV relative to the crystal axes. The map contains significant [100] (red) and [001] (blue) components, as expected. It also shows WBVs close to the [010] direction which are greenish and easily noticed in the map (top right and bottom left).

Remapping at a higher resolution of one of these regions (Figure 1c inset) gives a more detailed WBV map highlighting linear green structures suggestive of subgrain walls, particularly the feature running bottom-left to top-right (detailed inset) in which the WBV is almost parallel to [010].

To quantify the proportion of b dislocations we applied simple statistics to the a, b and c components of the WBV (see Supporting Information S1). These show that there is a 17% [010] contribution to the overall GND population in the main map and 19% in the inset map. These numbers are derived from the quantities V' which are proxies for densities of dislocations with different Burgers vectors (Table S1 in Supporting Information S1). Though they have the same units as dislocation density (m^{-2}), these are not true densities of GNDs, because the WBV is influenced by dislocation line orientations and GND density is step size dependent (Ruggles et al., 2016). Different densities would be calculated if mixed Burgers vectors were present, but the comparative magnitudes of proxies indicate a considerable proportion of b dislocations.

3.2. Confirming b Dislocations Using TEM

WBV calculations are based on very few assumptions but, given the repeated assertions that b dislocations are rare, we took a closer look. Targeting the linear green feature and guided by the previous BSE and EBSD images,

Focused Ion Beam (FIB) milling was used to remove an electron transparent wafer of material (Figures 1d–1g). A composite bright field TEM image (Figure 1h) shows NE-SW walls which can be lined up with the features in the WBV map from Figure 1c insets. We homed in on the wall of interest, using that image, and gathered SPED patterns. From these we made a VBF image of an area of interest (Figure 1i) and indexed the diffraction patterns to quantify crystal orientations. Orientation gradients run parallel to [010] and the Euler angle map (Figure 1j) confirms this, showing that misorientation axes (arrows on transect on map) run subparallel to [100]. The structure is in accord with a tilt boundary, with edge dislocation lines parallel to [100] and Burgers vectors parallel to [010]. We sought to confirm this with atomic scale images, but here the edge dislocation lines are at a low angle to the wafer and not ideal for views parallel to them, so we searched nearby. We used a composite LAADF image (Figure 1k); the gold “x” shows a reference point there and in Figure 1h as our guide. We magnified the area of interest in Figures 1l and 1m). The larger box in Figure 1m shows the location of Figure 1i; the smaller box shows an ENE linear structure with a hook on the right end, oriented SSE, parallel to [010] and imaged in ADF in Figure 1n.

In Figure 2a (enlarged from Figure 1n) the brightest spots are interpreted as follows. We overlaid a model of 4 unit cells viewed down the [001] axis (Figure 2c). In olivine there are two distinct cation sites, with M1 arranged in columns parallel to [001], in contrast to M2 which all have different positions along the [100] and [010] axes. So, the M1 columns have twice the projected “atomic density” (atoms are just $c/2 = 3 \text{ \AA}$ apart) and there is more scattering from them. In addition Fe is predicted to partition onto M1 (Gholamimahmoodabadi et al., 2024) increasing the average M1 atomic number and further enhancing brightness. M2 sites are in fact almost invisible in the simulated ADF image for a third reason: there is destructive interference (see SI for details). In the middle of the field of view there is blurring in an elongate feature; we demonstrate this is a screw dislocation with a [010] component using two features.

First, the traces of (020) planes indicated in red in Figure 2a are offset by $\frac{1}{4}[010]$ passing across the defect, and returning to the starting side the total offset is $\frac{1}{2}[010]$, suggesting this is a partial screw dislocation with Burgers vector $\frac{1}{2}[010]$ subparallel to the wafer. A Burgers circuit drawn on the bright sites shows a [100] edge dislocation is also present, with the extra half plane shown in blue, but this cannot explain the offsets parallel to [010].

Secondly, the center of the blurred feature shows a herringbone pattern of gray streaks instead of bright end-on M1 columns. We deduce these are rotated M1 columns. If this were due to an edge dislocation the rotation and blurring would be perpendicular to the dislocation line, so we infer a component of displacement parallel to [010] is involved. To understand the herringbone pattern further we built a model of M1 positions, using the unit cell and the displacement field around a screw dislocation in an infinite isotropic solid. We then simulated an “optical section” (Yang et al., 2015) including atoms through a limited thickness of the foil. In a classic screw dislocation, all atomic displacements are parallel to the dislocation line and a view down [001] would show M1 columns apparently parallel to the dislocation line, not herringbone. We explored several explanations (see Supporting Information S1) and the one which best matches observations is a local elastic rotational strain close to the dislocation, due to the nearby free surface and/or the [100] edge dislocation modifying the stress and strain fields that are expected in an infinite solid. For screw dislocations running along the axes of cylinders and discs, such fields have been predicted (Eshelby, 1958; Eshelby & Stroh, 1951) and observed (Ugarte et al., 2019) to have a strong rotational component (the “Eshelby twist”), and similarly for oblique screws near interfaces (Yoffe, 1961). We empirically modified our model to include a local elastic twist and find satisfactory agreement with the ADF (Figure 2b). The trace of the [010] screw shows that it deflects to the right in the bottom right of Figure 2a; it will there have an edge component in accord with the nearby structure in Figures 1i and 1j.

In summary the picture here is we have a $\frac{1}{2}[010]$ partial screw dislocation running parallel to the wafer, and a [100] edge with line running parallel to [001]. This is a complex geometry, but the WBV map guided us toward the [010] component, maximizing our chances of locating and identifying these features.

3.3. b Dislocations Are Present in Significant Quantities

WBV calculations involve few assumptions. It is assumed that elastic strains are small, and it is difficult to conceive how elastic strain could give such sharp distortions such as in Figure 1e, and distortions which by chance fit with crystallographic directions. Nevertheless, we validated the prediction of b dislocations using TEM. 17% of the dislocations in our sample may have Burgers vectors with a [010] component, in terms of dislocation density proxy (Table S1 in Supporting Information S1). Apparent GND density is dependent on step size

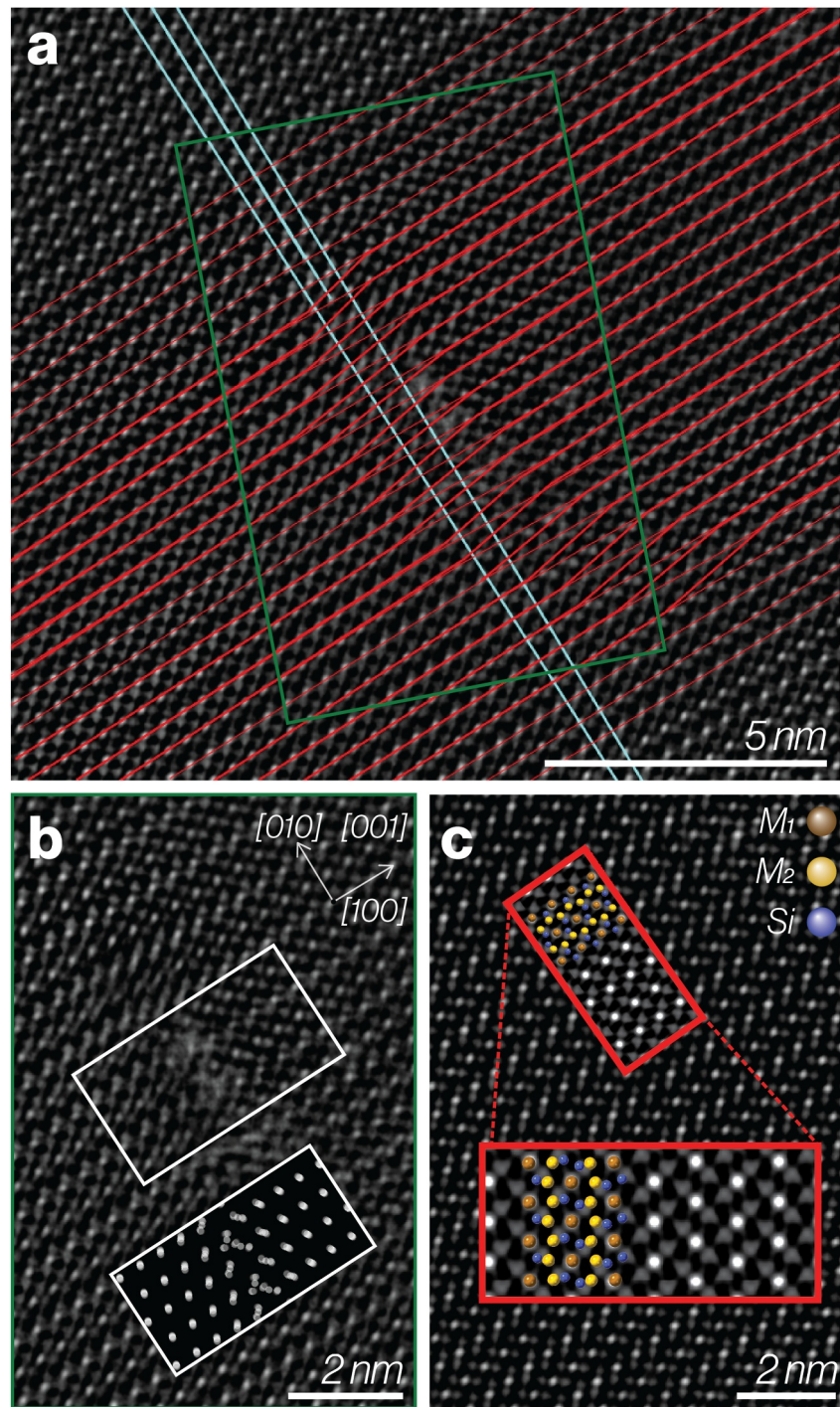


Figure 2. Atomic resolution STEM characterization of b dislocations in olivine. (a) Atomic-resolution ADF-STEM image of b dislocation in olivine with (020) planes highlighted in red and extra half plane related to a separate edge dislocation in blue. (b) Magnified image of central part and model of M1 positions in the dislocation inset at bottom, matching the white rectangle. (c) Atomic-resolution image of defect-free olivine with (insets) atomic columns identified using known positions and simulated ADF image.

(Ruggles et al., 2016) and there are some stereological assumptions behind linking the proxy GND density, but these issues affect all the dislocations in similar ways, so we conclude there is a substantial proportion of b dislocations; it is impossible to explain the observations without it. Calculations like ours would quickly give

insights into the proportions of dislocations with [100], [010] and [001] Burgers vectors in any olivine sample. Other methods to deduce GND densities from EBSD measurements of lattice curvature (Wilkinson et al., 2006) involve assumptions of dislocation line energies which may be set very high for [010] dislocations assuming they are “fictitious” (Wallis et al., 2016). Their presence may thus be obfuscated. Our method, based only on the geometric link between lattice curvature and GNDs, is free from such assumptions. More established methods for deducing dislocation types and hence slip systems using misorientation axes of subgrain walls tend to assume the walls have tilt character (Lloyd et al., 1997), again a restrictive assumption the WBV approach avoids.

3.4. b Dislocations Are Mobile

Immobile dislocations can form during crystal growth (Gardner et al., 2021; Spruzeniece et al., 2017) but our experiment did not involve any olivine phase transformations and the original grains are recognizable, so the b dislocations are due to deformation. Figure 1c shows that b dislocations occur in part in linear features which are qualitatively similar in form to those formed by [100] and [001] dislocations which are subgrain walls (see Figure S2 in Supporting Information S1). Subgrain walls are formed by glide and climb of dislocations (Section 11.1.1 of Mittemeijer (2021)) so we deduce [010] dislocations in this sample must be doing both. This agrees with inferences of b slip from direct observations of dislocations in experiments (Ando et al., 1997) and from natural CPOs (S. Jung et al., 2014; Skemer et al., 2006). The proportions of dislocation densities (Table S1 in Supporting Information S1) agree with assessments that the dislocation line energy is higher for [010] dislocations than for the other directions and suggests b slip is “not favored” (Karato et al., 2008) but may still contribute. The relative line energy of [010] and [001] dislocations is $(b/c)^2 = 2.9$ —we suggest that this ratio is not big enough to totally inhibit b slip. The mobility of the b dislocations contradicts the assertion that there is “no possibility of shear along [010]” and when observed that they are “geometrically necessary dislocations rather than gliding dislocations” (Demouchy, 2021). That phrase does not indicate how those authors think they formed, and, in any case, mobile dislocations can remain as GNDs after deformation. Moreover b dislocations dissociate into partials in nature (Fujino et al., 1993) and simulations (Furstoss et al., 2024), reducing their energy but not disabling glide or climb. Our work shows that claims about the abundance and mobility of b dislocations merit revisiting.

4. Discussion

We suggest that natural and experimentally deformed olivine must have a range of b dislocation contents: it seems unlikely that all samples studied to date have content <1% except ours with 17%, especially when b slip has been proposed to explain some CPOs (Table 1). Our sample was deformed at ~ 2 GPa and $850 \pm 50^\circ\text{C}$, within range of subduction zone conditions (van Keken & Wilson, 2023) so has relevance at actual mantle conditions.

The imposed strain in the experiment was uniaxial but Fig. 1(a) shows that the olivine was substantially extruded and since the cut was made through the middle of the sample and the map is off axis, we are looking at the XZ plane of deformation. In accord with this the CPO (Figure S1 in Supporting Information S1) is not uniaxial; it shows the (010) plane mainly perpendicular to the shortening direction and [001] parallel to the shear involved in extruding the olivine, resembling a type B fabric (H. Jung et al., 2006), in accord with the high stress (>0.5 GPa) in our experiment. This suggests (010)[001] as the main slip system, but our method shows there are slightly more [100] dislocations than [001] (Table S1 in Supporting Information S1), and roughly half as many b dislocations. The abundance of GNDs does not necessarily link directly to slip system efficiency but these new observations could give fresh insights into olivine slip systems. Slip parallel to b has not left an obvious signature in the CPO but may have affected strength. Samples with [010] parallel to lineation (X) are candidates for b slip and WBV investigation could help clarify current uncertainty in how they formed (H. Jung, 2011).

The asserted lack of b slip in olivine gives a problem with accommodating arbitrary imposed strain because the slip systems are too restricted to satisfy von Mises criterion (Paterson, 2013). Different orientations in a polycrystal can assist, as can grain boundary sliding (Ohuchi et al., 2015); we suggest that b slip should be considered as another accommodation mechanism and may need to be included in crystal plasticity models.

Stress levels may play a role in promoting b slip. Stress cannot be measured directly in the mantle, but models predict stresses of 100s of MPa in for example, slab bending (Buffett & Becker, 2012; Cizkova et al., 2007) or because of volume changes in the Transition Zone where olivine transforms to or from more dense minerals (Devaux et al., 2000). Table 1 shows that pressure may also have a significant effect on b slip as suggested by Ando et al. (1997). They suggested 1 GPa marks a minimum pressure at which substantial [010] slip can occur,

but that allows for its action through the entire upper mantle (1–14 GPa). Future work could build up a statistical picture of b slip in olivine and the effect of stress, pressure (with possible geobarometry) and temperature on its activity and abundance, hence allowing those parameters to be constrained in a new way even if overall strength is not greatly affected.

In summary our work is significant for these reasons.

- The approach of determining slip systems from CPOs may benefit from direct information on GNDs gathered by our method.
- Mobile b dislocations may be present in sufficient numbers to influence the rheology of olivine and fulfill the von Mises criterion.
- Their presence will be controlled by a combination of stress, pressure and temperature so they would provide valuable new information from natural samples on those conditions in the upper Earth.
- The WBV method is easily applied to EBSD maps from other minerals to better understand their deformation.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

EBSD data for the main and inset maps are in Wheeler et al. (2025). Software supporting this research is available in the “Dislocation Analysis” module of Aztec Crystal from Oxford Instruments. It is available as Matlab closed source code (“Crystalscape”) for academic use only; a legal agreement with Oxford Instruments excludes commercial use. Academic researchers can gain access on request via Zenodo (Wheeler, 2025).

Acknowledgments

J. Wheeler and S. Hunt were funded by NERC Grant NE/V018477/1. Y. Li acknowledges technical support from David Hall and financial support by the EPSRC (Grant EP/S028978/1 and EP/V053183/1). TEM and FIB access were supported by the Henry Royce Institute for Advanced Materials, funded through EPSRC Grants EP/R00661X/1, EP/S019367/1, EP/P025021/1 and EP/P025498/1, and further EPSRC Grant EP/S021531/1. We thank H. Jung and an anonymous reviewer for their inputs and F. Capitanio and M. Ramil for their editorial contributions.

References

- Ando, J.-I., Irifune, T., Takeshita, T., & Fujino, K. (1997). Evaluation of the non-hydrostatic stress produced in a multi-anvil high pressure apparatus. *Physics and Chemistry of Minerals*, 24(2), 139–148. <https://doi.org/10.1007/s002690050027>
- Ashby, M. F. (1970). Deformation of plastically non-homogeneous materials. *Philosophical Magazine*, 21(170), 399–424. <https://doi.org/10.1080/14786437008238426>
- Becker, T. W. (2006). On the effect of temperature and strain-rate dependent viscosity on global mantle flow, net rotation, and plate-driving forces. *Geophysical Journal International*, 167(2), 943–957. <https://doi.org/10.1111/j.1365-246X.2006.03172.x>
- Blackman, D. K., & Kendall, J. M. (2002). Seismic anisotropy in the upper mantle 2. Predictions for current plate boundary flow models. *Geochemistry, Geophysics, Geosystems*, 3(9). <https://doi.org/10.1029/2001GC000247>
- Boland, J. N., & Buiskool Toxopeus, J. M. A. (1977). Dislocation deformation mechanisms in periodotite xenoliths in kimberlites. *Contributions to Mineralogy and Petrology*, 60(1), 17–30. <https://doi.org/10.1007/bf00372848>
- Boland, J. N., & Liebermann, R. C. (1983). Mechanism of the olivine to spinel phase transformation in Ni_2SiO_4 . *Geophysical Research Letters*, 10(1), 87–90. <https://doi.org/10.1029/GL010i001p00087>
- Buffett, B. A., & Becker, T. W. (2012). Bending stress and dissipation in subducted lithosphere. *Journal of Geophysical Research*, 117(B5). <https://doi.org/10.1029/2012jb009205>
- Cizkova, H., van Hunen, J., & van den Berg, A. (2007). Stress distribution within subducting slabs and their deformation in the transition zone. *Physics of the Earth and Planetary Interiors*, 161(3–4), 202–214. <https://doi.org/10.1016/j.pepi.2007.02.002>
- Demouchy, S. (2021). Defects in olivine. *European Journal of Mineralogy*, 33(3), 249–282. <https://doi.org/10.5194/ejm-33-249-2021>
- Demouchy, S., Mussi, A., Weidner, T., Gardés, E., & Cordier, P. (2024). Dislocations in naturally deformed olivine: Example of a mylonitic peridotite. *Physics of the Earth and Planetary Interiors*, 346, 107125. <https://doi.org/10.1016/j.pepi.2023.107125>
- Devaux, J., Fleitout, L., Schubert, G., & Anderson, C. (2000). Stresses in a subducting slab in the presence of a metastable olivine wedge. *Journal of Geophysical Research*, 105, 13365–13373.
- Durham, W. B., & Goetze, C. (1977). Plastic flow of oriented single-crystals of olivine. 1. Mechanical data. *Journal of Geophysical Research*, 82(36), 5737–5753. <https://doi.org/10.1029/jb082i036p05737>
- Eshelby, J. D. (1958). The twist in a crystal whisker containing a dislocation. *Philosophical Magazine*, 3(29), 440–447. <https://doi.org/10.1080/14786435808244565>
- Eshelby, J. D., & Stroh, A. N. (1951). Dislocations in thin plates. *Philosophical Magazine*, 42(335), 1401–1405. <https://doi.org/10.1080/14786445108560958>
- Fujino, K., Nakazaki, H., Momoi, H., Karato, S., & Kohlstedt, D. L. (1993). TEM observation of dissociated dislocations with $b = 010$ in naturally deformed olivine. *Physics of the Earth and Planetary Interiors*, 78(1–2), 131–137. [https://doi.org/10.1016/0031-9201\(93\)90089-r](https://doi.org/10.1016/0031-9201(93)90089-r)
- Furstoss, J., Hirel, P., Carrez, P., Gouriet, K., Meko-Fotso, V., & Cordier, P. (2024). Structures and energies of twist grain boundaries in Mg_2SiO_4 forsterite. *Computational Materials Science*, 233, 112768. <https://doi.org/10.1016/j.commatsci.2023.112768>
- Gardner, J., Wheeler, J., & Mariani, E. (2021). Interactions between deformation and dissolution-precipitation reactions in feldspar at greenschist facies. *Lithos*, 396–397.
- Gholamimahmoodabadi, Z., Mondal, R., Chatterjee, S., & Saha-Dasgupta, T. (2024). Can increased pressure induce cross-over of Fe partitioning in olivine under upper mantle P–T conditions? *Journal of Earth System Science*, 133(4), 194. <https://doi.org/10.1007/s12040-024-02390-4>
- Hirth, J. P., & Lothe, J. (1982). *Theory of dislocations* (1st ed.). Malabar, FL: Krieger Publishing Company.

- Hunt, S. A., Weidner, D. J., McCormack, R. J., Whitaker, M. L., Bailey, E., Li, L., et al. (2014). Deformation T-Cup: A new multi-anvil apparatus for controlled strain-rate deformation experiments at pressures above 18 GPa. *Review of Scientific Instruments*, 85(8), 085103. <https://doi.org/10.1063/1.4891338>
- Hunt, S. A., Whitaker, M. L., Bailey, E., Mariani, E., Stan, C. V., & Dobson, D. P. (2019). An experimental investigation of the relative strength of the Silica polymorphs quartz, coesite, and stishovite. *Geochemistry, Geophysics, Geosystems*, 20(4), 1975–1989. <https://doi.org/10.1029/2018gc007842>
- Jung, H. (2011). Seismic anisotropy produced by serpentine in mantle wedge. *Earth and Planetary Science Letters*, 307(3), 535–543. <https://doi.org/10.1016/j.epsl.2011.05.041>
- Jung, H., Katayama, I., Jiang, Z., Hiraga, I., & Karato, S. (2006). Effect of water and stress on the lattice-preferred orientation of olivine. *Tectonophysics*, 421(1–2), 1–22. <https://doi.org/10.1016/j.tecto.2006.02.011>
- Jung, S., Jung, H., & Austrheim, H. (2014). Characterization of olivine fabrics and mylonite in the presence of fluid and implications for seismic anisotropy and shear localization. *Earth Planets and Space*, 66(1), 46. <https://doi.org/10.1186/1880-5981-66-46>
- Karato, S., Jung, H., Katayama, I., & Skemer, P. (2008). Geodynamic significance of seismic anisotropy of the upper mantle: New insights from laboratory studies. *Annual Review of Earth and Planetary Sciences*, 36(1), 59–95. <https://doi.org/10.1146/annurev.earth.36.031207.124120>
- Kirkland, E. J. (2020). *Advanced computing in electron microscopy by earl J. Kirkland* (3rd 2020). Cham: Springer International Publishing.
- Lloyd, G. E., Farmer, A. B., & Mainprice, D. (1997). Misorientation analysis and the formation and orientation of subgrain and grain boundaries. *Tectonophysics*, 279(1–4), 55–78. [https://doi.org/10.1016/s0040-1951\(97\)00115-7](https://doi.org/10.1016/s0040-1951(97)00115-7)
- Mittemeijer, E. J. (2021). *Fundamentals of materials science: The microstructure-property relationship using metals as model systems/eric J. Mittemeijer* (2nd ed.). Cham, Switzerland: Springer.
- Ohuchi, T., Kawazoe, T., Higo, Y., Funakoshi, K.-I., Suzuki, A., Kikegawa, T., & Irifune, T. (2015). Dislocation-accommodated grain boundary sliding as the major deformation mechanism of olivine in the Earth's upper mantle. *Science Advances*, 1(9), e1500360. <https://doi.org/10.1126/sciadv.1500360>
- Palmer, D. C. (2014). *CrystalMaker: crystalMakerSoftware Ltd.* Begbroke, Oxfordshire, England.
- Paterson, M. S. (2013). *Materials science for structural geology*. Springer.
- Phakey, P., Dollinger, G., & Christie, J. (1972). Transmission electron microscopy of experimentally deformed olivine crystals. In *Flow and fracture of rocks* (pp. 117–138).
- Prior, D. J., Mariani, E., & Wheeler, J. (2009). EBSD in the Earth sciences: Applications, common practice and challenges. In A. J. Schwartz, M. Kumar, B. L. Adams, & D. P. Field (Eds.), *Electron backscatter diffraction in materials science* (pp. 345–357). Springer.
- Ruggles, T. J., Rampton, T. M., Khosravani, A., & Fullwood, D. T. (2016). The effect of length scale on the determination of geometrically necessary dislocations via EBSD continuum dislocation microscopy. *Ultramicroscopy*, 164, 1–10. <https://doi.org/10.1016/j.ultramic.2016.03.003>
- Skemer, P., Katayama, I., & Karato, S.-I. (2006). Deformation fabrics of the Cima di Gagnone peridotite massif, Central Alps, Switzerland: Evidence of deformation at low temperatures in the presence of water. *Contributions to Mineralogy and Petrology*, 152(1), 43–51. <https://doi.org/10.1007/s00410-006-0093-4>
- Spruzeniece, L., Piazzolo, S., & Maynard-Casely, H. E. (2017). Deformation-resembling microstructure created by fluid-mediated dissolution-precipitation reactions. *Nature Communications*, 8(1), 14032. <https://doi.org/10.1038/ncomms14032>
- Thieme, M., Demouchy, S., Mainprice, D., Barou, F., & Cordier, P. (2018). Stress evolution and associated microstructure during transient creep of olivine at 1000–1200 degrees C. *Physics of the Earth and Planetary Interiors*, 278, 34–46. <https://doi.org/10.1016/j.pepi.2018.03.002>
- Tielke, J. A., Zimmerman, M. E., & Kohlstedt, D. L. (2017). Hydrolytic weakening in olivine single crystals. *Journal of Geophysical Research-Solid Earth*, 122(5), 3465–3479. <https://doi.org/10.1002/2017jb014004>
- Ugarte, D., Tizei, L. H. G., Cotta, M. A., Ducati, C., Midgley, P. A., & Eggeman, A. S. (2019). Analysis of structural distortion in eshelby twisted InP nanowires by scanning precession electron diffraction. *Nano Research*, 12(4), 939–946. <https://doi.org/10.1007/s12274-019-2328-5>
- van Keken, P. E., & Wilson, C. R. (2023). An introductory review of the thermal structure of subduction zones: III-comparison between models and observations. *Progress in Earth and Planetary Science*, 10(1), 57. <https://doi.org/10.1186/s40645-023-00589-5>
- Wallis, D., Hansen, L. N., Ben Britton, T., & Wilkinson, A. J. (2016). Geometrically necessary dislocation densities in olivine obtained using high-angular resolution electron backscatter diffraction. *Ultramicroscopy*, 168, 34–45. <https://doi.org/10.1016/j.ultramic.2016.06.002>
- Wheeler, J. (2025). *Crystalscape (version 2.5.0) [Software]*. Zenodo. <https://doi.org/10.5281/zenodo.15465607>
- Wheeler, J., Hunt, S. A., Eggeman, A. S., Donoghue, J., Gholinia, A., Li, Y., et al. (2025). Electron backscatter diffraction CTF files [Dataset]. Zenodo. <https://doi.org/10.5281/zenodo.17493851>
- Wheeler, J., Piazzolo, S., Prior, D. J., Trimby, P. W., & Tielke, J. A. (2024). Using crystal lattice distortion data for geological investigations: The weighted burgers vector method. *Journal of Structural Geology*, 179, 105040. <https://doi.org/10.1016/j.jsg.2023.105040>
- Whitehouse, P. L. (2018). Glacial isostatic adjustment modelling: Historical perspectives, recent advances, and future directions. *Earth Surface Dynamics*, 6(2), 401–429. <https://doi.org/10.5194/esurf-6-401-2018>
- Wilkinson, A. J., Meaden, G., & Dingley, D. J. (2006). High resolution mapping of strains and rotations using electron backscatter diffraction. *Materials Science and Technology*, 22(11), 1271–1278. <https://doi.org/10.1179/174328406x130966>
- Yang, H., Lozano, J. G., Pennycook, T. J., Jones, L., Hirsch, P. B., & Nellist, P. D. (2015). Imaging screw dislocations at atomic resolution by aberration-corrected electron optical sectioning. *Nature Communications*, 6(1), 7266. <https://doi.org/10.1038/ncomms8266>
- Yoffe, E. H. (1961). A dislocation at a free surface. *Philosophical Magazine*, 6(69), 1147–1155. <https://doi.org/10.1080/14786436108239675>
- Zhang, J., Li, B., Utsumi, W., & Liebermann, R. C. (1996). In situ X-ray observations of the coesite-stishovite transition: Reversed phase boundary and kinetics. *Physics and Chemistry of Minerals*, 23(1), 1–10. <https://doi.org/10.1007/BF00202987>

References From the Supporting Information

- De Graef, M. (2020). A dictionary indexing approach for EBSD. *IOP Conference Series: Materials Science and Engineering*, 891(1), 012009. <https://doi.org/10.1088/1757-899x/891/1/012009>
- Gröger, R., Dudeck, K. J., Nellist, P. D., Vitek, V., Hirsch, P. B., & Cockayne, D. J. H. (2011). Effect of Eshelby twist on core structure of screw dislocations in molybdenum: Atomic structure and electron microscope image simulations. *Philosophical Magazine*, 91(18), 2364–2381. <https://doi.org/10.1080/14786435.2011.562474>
- Haider, M., Uhlemann, S., Schwan, E., Rose, H., Kabius, B., & Urban, K. (1998). Electron microscopy image enhanced. *Nature*, 392(6678), 768–769. <https://doi.org/10.1038/33823>

- Nolze, G., Jurgens, M., Olbricht, J., & Winkelmann, A. (2018). Improving the precision of orientation measurements from technical materials via EBSD pattern matching. *Acta Materialia*, 159, 408–415. <https://doi.org/10.1016/j.actamat.2018.08.028>
- Pennycook, S. J., & Jesson, D. E. (1991). High-resolution z-contrast imaging of crystals. *Ultramicroscopy*, 37(1–4), 14–38. [https://doi.org/10.1016/0304-3991\(91\)90004-p](https://doi.org/10.1016/0304-3991(91)90004-p)
- Prior, D. J., Trimby, P. W., Weber, U. D., & Dingley, D. J. (1996). Orientation contrast imaging of microstructures in rocks using foreshatter detectors in the scanning electron microscope. *Mineralogical Magazine*, 60(403), 859–869. <https://doi.org/10.1180/minmag.1996.060.403.01>
- Rauch, E. F., Portillo, J., Nicolopoulos, S., Bultreys, D., Rouvimov, S., & Moeck, P. (2010). Automated nanocrystal orientation and phase mapping in the transmission electron microscope on the basis of precession electron diffraction. *Zeitschrift Fur Kristallographie-Crystalline Materials*, 225(2–3), 103–109. <https://doi.org/10.1524/zkri.2010.1205>
- Rauch, E. F., & Véron, M. (2014). Virtual dark-field images reconstructed from electron diffraction patterns. *The European Physical Journal - Applied Physics*, 66(1), 10701. <https://doi.org/10.1051/epjap/2014130556>
- Rauch, E. F., Veron, M., Portillo, J., Bultreys, D., Maniette, Y., & Nicolopoulos, S. (2008). Automatic crystal orientation and phase mapping in TEM by precession diffraction. *Microscopy and Analysis*, 22, S5–S8.
- Wheeler, J., Mariani, E., Piazzolo, S., Prior, D. J., Trimby, P., & Drury, M. R. (2009). The weighted burgers vector: A new quantity for constraining dislocation densities and types using electron backscatter diffraction on 2D sections through crystalline materials. *Journal of Microscopy*, 233(3), 482–494. <https://doi.org/10.1111/j.1365-2818.2009.03136.x>
- Winkelmann, A., Jablon, B. M., Tong, V. S., Trager-cowan, C., & Mingard, K. P. (2020). Improving EBSD precision by orientation refinement with full pattern matching. *Journal of Microscopy*, 277(2), 79–92. <https://doi.org/10.1111/jmi.12870>